

Business Performance Analytics

&

Predictive Modeling

(Formula 1 Data as a Business Simulation)

1. Business Problem Definition

A high-performance organization aims to analyze operational efficiency, financial impact, and performance consistency across business units to improve outcomes and support strategic decision-making.

Although the data originates from Formula 1 racing, the project is designed as a business-generic performance analytics framework, where drivers, teams, and races simulate employees, departments, and operational cycles.

This approach demonstrates how advanced analytics techniques used in sports can be directly applied to corporate performance management, financial analysis, and strategic planning.

2. Stakeholder Analysis

Stakeholder	Primary Concerns
Executives	ROI, overall performance, predictability
Finance	Cost efficiency, performance per unit
Operations	Process optimization, reliability
Strategy	Long-term trends, consistency, risk

3. Key Performance Indicators (KPIs)

KPI	Business Meaning
Performance Index	Output effectiveness
Efficiency Score	Output vs effort
Consistency Index	Stability over time
ROI Proxy	Performance per unit of resource

4. Data Collection

4.1 Data Source

Data was collected from the Ergast Developer API, a public Formula 1 data service.

Due to network restrictions, used a keggles dataset which is publicly available mirror of the official Ergast dataset was used. And did all the analysis task over the past 10 years only.

4.2 Tools Used

- Python
- SSMS
- pandas
- Jupyter Notebook
- Power BI
- Visual Studio

4.3 Datasets Collected

Dataset	Rows	Columns	Description
races	1,149	18	Race events
drivers	864	9	Drivers
constructors	212	5	Teams
results	27,238	18	Performance per race
lap_times	12,192	7	Timing & efficiency data

4.4 Technical Improvement

To improve portability and avoid file path issues, the project structure was updated to use:

- `pathlib.Path`
- Centralized project root directory

Benefit:

- Platform-independent
- Easier execution via terminal or VS Code
- Cleaner project structure

5. Data Cleaning & Preparation

5.1 Tasks Performed

- Handled missing values
- Normalized date and numeric formats
- Removed duplicates
- Merged datasets using primary & foreign keys

5.2 Tools

- Python (Pandas)
- Excel

6. Exploratory Data Analysis (EDA)

6.1 Business Questions Answered

- Which business units consistently outperform?
- What operational factors drive results?
- Is higher variance linked to higher performance?
- Which units are reliable vs high-risk?

6.2 Visual Analysis

- Time-series trends
- Correlation matrices
- Performance distributions

6.3 Tools

- Python
- Matplotlib
- Seaborn

7. Feature Engineering

7.1 Purpose

To transform raw operational data into **decision-ready performance indicators**.

7.2 New Metrics Created

- Performance volatility
- Efficiency ratio
- Consistency score
- Long-term performance averages

7.3 Tools

- Python

8. SQL Business Analysis

8.1 Implementation

- Loaded processed datasets into:
SQL Server Management

8.2 Business Queries

- Benchmarking top vs average performers
- Trend analysis across seasons
- Segmentation by performance tiers

9. Power BI Dashboard (BI Visualization Layer)

9.1 Dashboard Pages

1. Executive Overview
2. Operational Efficiency
3. Financial Impact
4. Predictive Insights

9.2 Tools

- Power BI
- DAX (basic measures)

9.3 Business Mapping

F1 Concept	Business Interpretation
Points	Performance Score
Reliability	Risk / Consistency
Season	Year / Quarter
Wins	High-value outcomes
DNFs	Failures / Attrition

10. Predictive Analytics

10.1 Models Implemented

Model Type	Business Use
Regression	Predict future performance
Classification	High vs low performer
Clustering	Strategy & behavior profiles

10.2 Tools

- Python
- Scikit-learn

11. Business Insights & Recommendations

11.1 Key Insights

- Consistency is a stronger predictor than peak performance
- High performers show lower volatility
- Team-level efficiency amplifies individual success

11.2 Financial Implications

- Resource allocation should prioritize consistent units
- Risk-adjusted performance yields better ROI

11.3 Strategic Actions

- Invest in reliability-focused teams
- Optimize operational efficiency over short-term wins
- Use predictive models for proactive decision-making